



LambdaProof Evidence Validation

Supported Evidence Types, Formats, and Validation Rules

Overview

LambdaProof uses a **multi-layered evidence validation system** that ensures every capability token, policy decision, and audit trail is backed by cryptographically verifiable evidence. The system supports both on-chain and off-chain validation, with hybrid post-quantum (PQ) readiness for future-proofing.^{[1][2]}

Evidence in LambdaProof serves three core purposes:

- **Authorization** - proving a subject has the right to perform an action
 - **Compliance** - demonstrating adherence to policy and regulatory requirements
 - **Auditability** - providing regulator-grade verification without exposing raw data
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Supported Evidence Types

Zero-Knowledge Proofs

Proof Type	Format	Use Case
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Groth16	<code>.zkey, .wasm, vk.json</code>	Identity attestation, policy evaluation, transaction authorization ^[1]
SNARK/Merkle Proofs	<code>snark-cid, merkle-proof-cid</code>	Archivum document certification, set membership ^[3]
ZK-KYC Proof	<code>proof + publicSignals</code>	Onboarding without exposing identifiers ^[1]
Screening Proof	Membership/non-membership proof	AML/sanctions verification without disclosure ^[1]
Recovery Proof	Groth16 over recovery circuit	Key replacement without PII exposure ^[1]

Attestations

Attestation Type	Format	Integrity Anchor
COSE/JWS Detached	<code>alg: Ed25519 or secp256k1</code>	Bank/HSM signing key ^[1]
Screening Attestation	JSON + bank signature	<code>accountCommitment, listVersion, outcome, ttl</code> ^[1]
Controls Attestation	Signed mapping	Evidence pointers for audit cycles ^[1]
ML-DSA (Post-Quantum)	<code>sig, sigpub</code> fields	Off-chain PQ receipts for hybrid security ^[4]

Receipts and Certificates

Receipt Type	Producer	Medium
ProofReceipt	Client/Indexer	JSON with <code>proofHash, vkHash, contract, block, event</code> ^[1]
ConsentReceipt	Client	JSON + user signature with <code>txParamsHash, policyId, nonce, expiresAt</code> ^[1]

ψ-Certificate	Archivum Validator	Ed25519-signed stability certificate with entropy/drift metrics ^[3]
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On-Chain Evidence

Evidence Type	Storage	Anchor
CustomerRoot	CircuitRegistry/MTPICore	Merkle root of KYC state ^[1]
AccountCommitment	MTPICore mapping	Chain state binding ^[1]
ProofVerified Event	Contract event	accountCommitment, txHash, policyId, deltaBound ^[1]
AuditTrailRoot	Periodic chain anchor	Rolling Merkle root over receipts ^[1]
Taproot Anchor	Bitcoin P2TR output	anchorroot = SHA256(merkleroot batchheaderhash) ^[4]

Evidence Formats

ProofReceipt Schema

```
{
  "type": "ProofReceipt",
  "proofHash": "0x...",
  "vkHash": "0x...",
  "contract": "0x...",
  "block": 12345,
  "event": "0x...",
  "issuedAt": 1704672000
}
```

ConsentReceipt Schema

```
{
  "type": "ConsentReceipt",
```

```
"txParamsHash": "0x...",
"policyId": "0x...",
"nonce": "0x...",
"expiresAt": 1704758400,
"sig": "0x..."
}
```

AccountBinding Schema

```
{
  "type": "AccountBinding",
  "userCommitment": "0x...",
  "accountCommitment": "0x...",
  "bankId": "BANKXYZ",
  "issuedAt": 1704672000,
  "bankSig": "0x..."
}
```

Archivum Object Evidence Schema

```
{
  "id": "CIDv1 or BLAKE3 multihash",
  "kind": "document|claim|tensor|operator|module|bundle",
  "factorization": {
    "primes": [
      {
        "pid": "p00000013",
        "eigenmode": 13,
        "payloaddigest": "blake3-256"
      }
    ]
  },
  "witnesses": ["proof-cid"],
  "proof": "snark-cid|merkle-proof-cid",
  "metrics": {
    "entropy": {"H_shannon": 1.42, "H_kolmogorov_est": 1.88},
    "drift": {"τ": 0.06}
  },
  "status": "pending|stable|tainted",
}
```

```
"certificate": "cid"
}
```

Validation Rules

Proof Validation

1. **Verifier Key Hash** - `vkHash` must match the circuit's pinned verification key^[1]
2. **Public Signal Ordering** - Must match the verifier ABI exactly^[1]
3. **PTAU Hash** - Must be pinned; deterministic builds required^[1]
4. **Client-Side Pre-Check** - Client verifies locally before any on-chain call^[1]

Attestation Validation

1. **Signature Algorithm** - Must use declared algorithm (Ed25519, secp256k1, ML-DSA)
2. **Signing Key** - Must trace to registered bank/HSM key
3. **Expiry** - TTL must not be exceeded at verification time
4. **Detached Mode** - COSE/JWS must be in detached signature mode for privacy^[1]

Receipt Validation

1. **Event Hash Match** - `eventHash` must match the on-chain event data
2. **Block Confirmation** - Block number must have required finality depth (d=3 mainnet, d=1 regtest)^[4]
3. **Integrity Chain** - Receipt hashes must form valid chain in `AuditTrailRoot`^[1]

Certificate Validation (ψ_{∞} -Stability)

Certificates are **revoked** when:^[3]

- Provenance withdrawal or signature revocation occurs
- New contradictory evidence with weight $> \delta$ appears
- Entropy spike: `H_shannon` $> \theta$ or `K_est` $> \kappa$

- Drift spike: $\text{embedding}_{\delta} > \varepsilon$ or $\text{edge}_{\delta} > \varepsilon$ after updates

Evidence Validation APIs

Verify Proof

POST /proofs/verify

Body: { "proof": {...}, "publicSignals": [...] } }

Response: { "status": "valid|invalid", "proofHash": "0x...", "vkHash": "0x..." } }

Issue Attestation

POST /attestations/issue

Body: { "type": "...", "payload": {...} } }

Response: { "attestation": {...}, "sig": "0x..." } }

Query Audit Receipt

GET /audit/receipts/{hash}

Response: ProofReceipt object

Verify Anchor Proof

POST /anchors/verify

Body: { "anchorProof": {...}, "root": "0x..." } }

Response: { "valid": true|false } }

Integrity Anchors Summary

Evidence Type	Integrity Anchor	Primary Consumers
KYCProof	vk.json hash on-chain	Bank, Verifier
Attestation	Bank signing key	Client, Auditor

CustomerRoot	Chain state event	Everyone
SpendProof	vk.json hash	Contract
ConsentReceipt	User signing key	Bank, Dispute
ProofReceipt	Event hash	Auditor, Regulator
AuditTrailRoot	Chain anchor (optional)	Regulator
Taproot Anchor	Bitcoin P2TR commitment	All validators

Constraints and Requirements

All evidence validation adheres to these core principles:^[1]

- **No raw identifiers** in any output—commitments only
- **Public signal ordering** must match verifier ABI
- **PTAU hash pinned**; vk/zkey hashes recorded
- **Client verifies locally** before any on-chain call
- **Drift bound (deltaBound)** enforced per policy—reject otherwise

Post-Quantum Readiness

ΛProof implements **hybrid PQ posture**.^[4]

- **On-chain**: Bitcoin L1 remains Schnorr/Taproot (no native PQ)
- **Off-chain**: Receipts signed with **ML-DSA** (CRYSTALS-Dilithium)
- **Verification path**: PQ signature → Merkle inclusion → AnchorProof → on-chain confirms existence

This approach ensures quantum resistance for evidence integrity while maintaining compatibility with current blockchain infrastructure.

Related Documentation

- **Banking Protocols** - Domain-specific evidence flows for KYC, transactions, AML
- **Healthcare Protocols** - Consent and data access evidence
- **Λ -Trace Architecture** - Audit layer and receipt management
- **Bitcoin Integration** - Taproot anchoring for evidence roots

Λ Proof Evidence Validation Specification v1.0

Public Document - January 2026

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1. 5.-Lproof-Banking-Protocols.pdf
 2. LProof-IP-Patent-Research.pdf
 3. Lp-Archivum.pdf
 4. Lproof-Bitcoin-Integration-Overview.pdf
 5. 2. Λ RootContract II & III - MTPI - NODE.js.pdf
 - a. Aproof Monorepo — Plain-language Overview V0.pdf
 - b. Λ -trace Overview.pdf
 - c. Aproof Healthcare Protocols V0.1.pdf
 - d. Adapps On Web4 - Developer Overview.pdf
 6. Building a Λ Dapp_ A Developer's Implementation Guide.pdf
 7. MTPI_Instructions.pdf
 8. MTPI RootContract.pdf
 9. MTPI_RootContract_Project.pdf
 10. Patent Search Results.pdf

11. The Λ Proof Architecture_ A Comprehensive Analysis of its Implications for Web4.pdf
12. The Λ Proof Architecture_ Web4.pdf
13. Verifiable, Privacy-Preserving Applications.pdf
14. Conscious_Sovereignty_Layer.pdf
15. Milestone-Vision-Timeframe.csv
16. Lets-further-develop-the-SAFE_-awesome-lets-turn.pdf
17. Ξ Constitution.pdf
 - e. Λ proof $\times$ Mtpi Lawful-by-design Integration — Implementation Plan.pdf